

**Jas Esperanza Hollis**  
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## **EDUCATION**

**Yale University**, New Haven, CT

Bachelor of Science, Earth and Planetary Science; GPA: 3.75

Expected May 2027

*Relevant Coursework:* Introductory Physics, Introductory Chemistry, Physics Laboratory Sequence, Introduction to Computing (C, C++, MATLAB), Partial and Ordinary Differential Equations, Linear Algebra, Fluid Mechanics, Applied Numerical Methods for Differential Equations (MATLAB), Weather and Climate Physics, Physical Oceanography, Geophysical Fluid Dynamics, Applied Statistical Methods

## **RESEARCH EXPERIENCE**

**Yale University, Earth and Planetary Science Department**

*First-Year STEM Summer Research Fellowship*, June – August 2024

Advisors: Mary-Louise Timmermans, Ashley Arroyo

I used observations and reanalysis data to 1) investigate the effect that cyclonic storms over the Arctic Ocean have on mixed-layer dissolved oxygen content in the Canada Basin, 2) assess how this effect is modulated by evolving sea ice extent.

**Massachusetts Institute of Technology, Earth, Atmospheric, and Planetary Science Department**

*MIT Summer Research Program*, June – August 2025

Advisor: Timothy Cronin

I worked with Dr. Cronin on a theoretical project to derive a simple analytic approximation of precipitation efficiency as a function of relative humidity, air temperature, and initial raindrop radius. I considered this question in the context of the sub-cloud layer under dry-adiabatic conditions and tested it in simulations of the exact equations.

**California Institute of Technology, Environmental Science and Engineering Department**

*WAVE Fellows Program*, June – August 2026

Advisors: Andrew Thompson, Sarah Zhang

I used ECCO Version 4, Release 5 estimate solutions to 1) quantify the contribution of snow accumulation on water mass transformation in the Southern Ocean, and 2) assess the effect that evolving sea ice extent has on this contribution.

## **PRESENTATIONS**

MSRP Summer Symposium, Cambridge, MA, August 2025. Hollis, Jas. “A Raindrop’s Journey: Approximating Precipitation Efficiency” (poster).

Caltech Summer Seminar Day, Pasadena, CA, August 2026. Hollis, Jas. “The Impact of Snow Accumulation on the Southern Ocean Overturning Circulation” (oral presentation).

## **TECHNICAL SKILLS**

**Computational:** C, C++, MATLAB, Python, Isca, ECCO, MITgcm, Machine Learning Engineering, Numerical Methods